Smayan Nandakumar

smayan.nandakumar@gmail.com | Personal Site | LinkedIn Page | GitHub Link

Professional Summary

Computer scientist specializing in quantum error correction, machine learning, quantum computing, optimization theory, and deep learning efficiency. Experienced in executing hardware-level quantum ablation studies on IBM processors, developing algebraic methods for quantum error correction, and translating classical NP-hard optimization problems into quantum formulations (QAOA, QUBO). Architected custom neural networks from scratch utilizing PyTorch and CUDA to accelerate training pipelines. Applied first-principles mathematics to complex algorithm development and deployment.

Education

Rensselaer Polytechnic Institute Troy, NY
Master of Science in Computer Science, magna cum laude *Aug. 2025 - May 2026*

- **GPA**: 3.84 / 4.00
- **Master's Project**: *Sailing the Noise Ocean: A Hardware Agnostic Noise Ablation Study of Quantum-Classical Neural Networks*

Rensselaer Polytechnic Institute Troy, NY
Bachelor of Science in Computer Science and Mathematics, magna cum laude *Aug. 2022 - Sep. 2025*

- **GPA**: 3.81 / 4.00

Relevant Coursework

Quantum & Theory: Distributed Quantum Computing (G), Approximation Algorithms (G), Design & Analysis of Algorithms, Quantum Physics I
Mathematics: Linear Algebra, Abstract Algebra, Intro to Topology, Mathematical Analysis I, Intro to Number Theory, Intermediate Logic, Linear and Conic Optimization (G)
Machine Learning: Reinforcement Learning (G), Machine Learning from Data, AI for Conservation (G)

Experience

Graduate Researcher Sep. 2025 - May 2026
Rensselaer Polytechnic Institute *Troy, NY*

- Conducted an ablation study to analyze the resilience of Hybrid Quantum-Classical Convolutional Neural Networks (QCQNNs) against 2 noise channels—Depolarizing and Thermal noise—on a 127-qubit IBM Eagle processor.
- Utilized Qiskit's Estimator to extract a 100% homogeneous median noise profile to demonstrate how elevated noise levels induce Barren Plateaus, flattening the loss landscape and preventing convergence.
- Trained and evaluated 12 distinct Noise models—running 3 iterations across a complete 4-configuration ablation matrix—to systematically isolate and quantify the impact of individual noise effects.

Undergraduate Mentor - Data Structures Jan. 2026 - May 2026
Rensselaer Polytechnic Institute *Troy, NY*

- Guided 50+ students through challenging memory management and foundational data structure concepts, actively assisting with real-time code debugging.
- Deconstructed 3+ complex topics—such as custom memory allocation, balanced search trees, and graph traversal—into clear, first-principles explanations to improve student comprehension.

Undergraduate Mentor - Intro To Algorithms Sep. 2024 - Dec. 2024
Rensselaer Polytechnic Institute *Troy, NY*

- Mentored 20+ students during weekly lab sessions, guiding them through complex algorithmic concepts and grading implementations to assess mathematical accuracy.
- Evaluated 20+ student solutions across 4+ core algorithmic paradigms—including memoization approaches, greedy algorithms, linear optimization, and polynomial-time reductions—providing rigorous feedback on time-space complexity and NP-hardness proofs.

Projects

Quantum Error Correction Pipeline | *Python, Qiskit, IBM Quantum* Sep. 2026 - Present

- Creating pipeline that will process any quantum circuit, and return a fault tolerant version with specified quantum stabilizer code.
- Utilizing symplectic linear algebra over $\mathbb{Z}_2$ to represent stabilizer generators.
- Authoring associated technical paper, detailing algorithms and rigorously proving relevant theorems.

Quantum Bin Packing via QAOA | *Python, Qiskit, IBM Quantum* Apr. 2026 - May 2026

- Developed a hybrid framework to solve the 0/1 Bin Packing problem by embedding a Knapsack solver into a Dantzig-Wolfe column generation loop, archiving the codebase in a public GitHub repository.
- Benchmarking performance on physical IBM hardware utilizing the Scholl dataset, down scaling instances to 6-10 items uniformly.

Neural Network From Scratch | *Python, NumPy* Nov. 2024 - Dec. 2024

- Built a fully functional neural network using only NumPy to classify handwritten digits from the 70000-image MNIST dataset.
- Implemented the entire mathematical pipeline, including forward propagation, activation functions, and loss function calculations, and achieved $> 80\%$ accuracy.

Replicated Key-Value Store with Paxos | *Python, UDP Sockets* Oct. 2024 - Nov. 2024

- Implemented a fault-tolerant, replicated key-value store across a 5 node cluster in Python using UDP socket programming, docker containers for simulation, and the Synod variant of the Paxos consensus algorithm.
- Enabled distributed nodes to reliably reach agreement on updates, ensuring consistency and availability in the presence of failures.

Technical Skills

Languages & Frameworks: Python, Java, PyTorch, CUDA, Qiskit, PennyLane, NumPy
Domains: Quantum Error Correction, Quantum Machine Learning, Optimization Theory, Graph Neural Networks, Distributed Algorithms, Computer Vision, Signal Processing
Developer Tools: Git, Docker

Awards

- *Dean's Honor List, RPI* — 2022, 2023, 2024
- *Math and Computer Science Department Award, EGHS* — 2022
- *Science Department Award, EGHS* — 2022

Personal Interests

I am a U.S. Citizen. I enjoy singing, minored in music in college, and was a secretary of Duly Noted RPI Acappella. I dabble in audio / sound engineering. I also enjoy playing various genres of board and video games, and read in my free time. I have volunteered in special needs summer camps, and have a special interest in *inclusion for all.*